

Lesson 22: Gauss-Markov Theorem and OLS Theory

BSTA 551: Statistical Inference: Bridging Estimation Theory to Linear Regression

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Where We've Been, and Where We're Going

Lessons 1–6: Point Estimation Theory

- Bias, variance, MSE
- MVUE: minimum variance among **all** unbiased estimators
- Cramér-Rao Lower Bound, Fisher Information
- Rao-Blackwell, Lehmann-Scheffé

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Lessons 7–21: Confidence intervals, hypothesis testing, bootstrap, nonparametric methods

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Today: How does all of that theory connect to **linear regression**, the crux of applied statistics?

Why This Matters

Your applied courses (regression, GLMs, mixed models) rely fundamentally on the optimality theory we have built. Today we make that connection explicit.

Today's Goals

1. Define the **simple linear regression (SLR) model** and the Gauss-Markov assumptions
2. Derive the **OLS estimators** from the principle of least squares
3. Prove OLS estimators are **linear** and **unbiased**
4. Derive their **exact variances**
5. State and prove the **Gauss-Markov Theorem**: OLS is BLUE
6. Connect BLUE $\rightarrow$ **MVUE** under the additional assumption of normality
7. Simulate to see the theorem in action with a **medical research example**

Textbook References

Topic	DBC	Notes
SLR model and OLS derivation	§12.1	Normal equations, $\hat{\beta}_0, \hat{\beta}_1$
Unbiasedness, $\text{Var}(\hat{\beta}_1)$	§12.2	Properties under GM assumptions
Gauss-Markov Theorem (BLUE)	§12.2	Stated in DBC, proof here
OLS = MLE under normality	§12.2	Connects to inference (t-tests)
Weighted least squares	§12.9	When homoscedasticity fails

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Notation

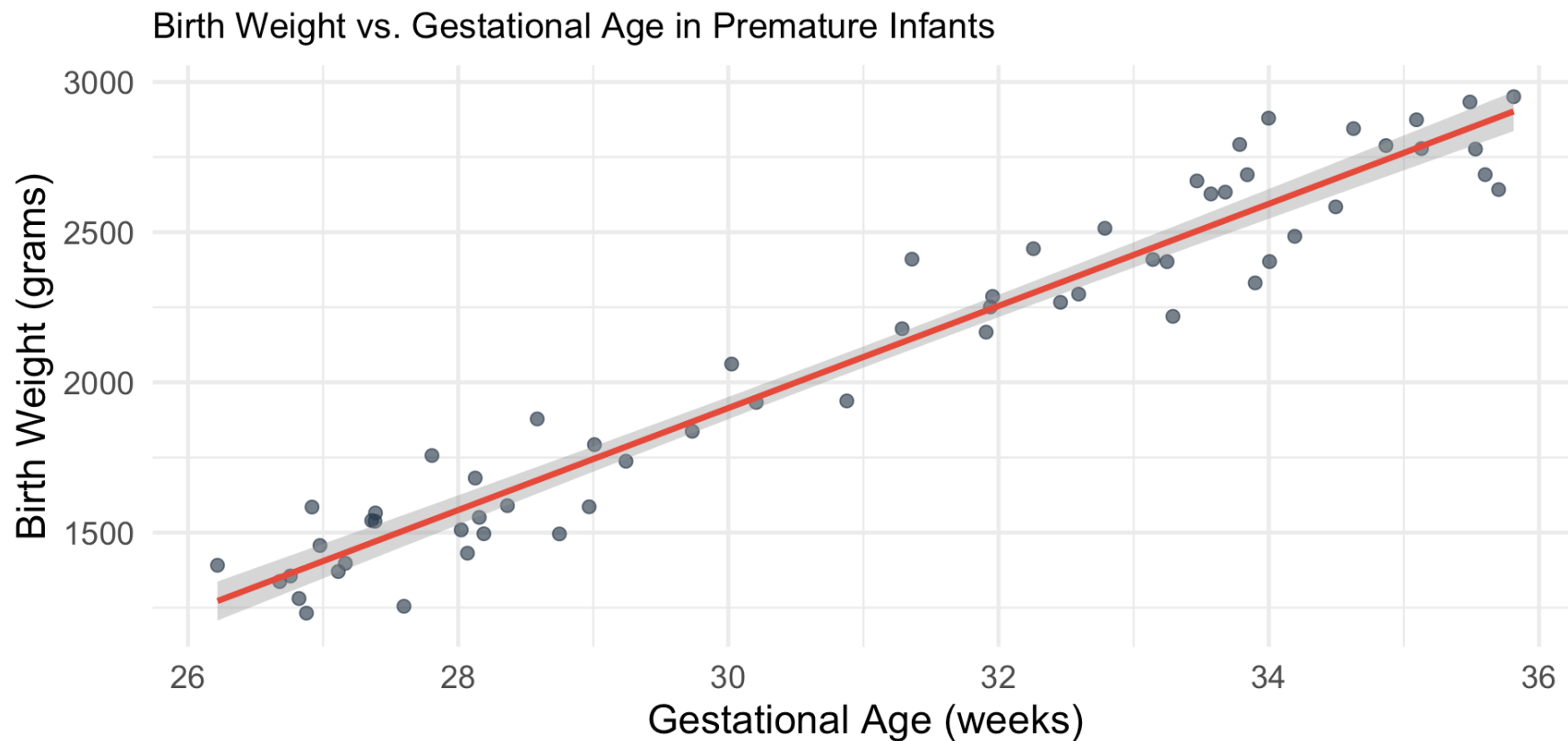
DBC uses β_0, β_1 for the **true** parameters and $\hat{\beta}_0, \hat{\beta}_1$ for the **OLS estimates**. We will use this convention throughout.

Part 1: The Simple Linear Regression Model

Motivating Example: Gestational Age and Birth Weight

A neonatologist wants to predict **birth weight** (grams) from **gestational age** (weeks) in premature infants.

► Code



Question: What is the “best” line through these data, and in what sense is it best?



The Simple Linear Regression Model

! Model (DBC §12.1)

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

where:

- Y_i = response (random variable) for the i -th observation
- x_i = predictor value (fixed, non-random), in a **fixed-design** setup
- β_0 = true intercept (unknown parameter)
- β_1 = true slope (unknown parameter)
- ε_i = random error term

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Notice: The x_i 's are **not random**; we treat them as known constants.

This is different from our earlier iid setup! The Y_i 's are **independent but not identically distributed**: each has a different mean $\beta_0 + \beta_1 x_i$.

The Gauss-Markov Theorem: Our Destination

Before examining the assumptions, let's state the theorem we are building toward.

! Gauss-Markov Theorem (preview)

Under four conditions on the error terms ε_i , the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **Best Linear Unbiased Estimators** (BLUE) of β_0 and β_1 .

That is, among all estimators that are (1) linear functions of $Y_1, \dots, Y_n$ and (2) unbiased, OLS has the **smallest variance**.

$$\text{Var}(\hat{\beta}_1^{\text{OLS}}) \leq \text{Var}(\tilde{\beta}_1) \quad \text{for every linear unbiased } \tilde{\beta}_1$$

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$$\text{Var}(\hat{\beta}_1^{\text{OLS}}) \leq \text{Var}(\tilde{\beta}_1) \quad \text{for every linear unbiased } \tilde{\beta}_1$$

Why does this matter? This is the regression analog of everything we built in Lessons 1–6: MVUE, CRLB, sufficiency. The four conditions (the **Gauss-Markov assumptions**) are exactly what makes OLS optimal. Understanding them tells you when OLS is trustworthy, and what to do when it is not.

The Gauss-Markov Assumptions

! Gauss-Markov (GM) Assumptions

1. **Linearity:** $E[Y_i] = \beta_0 + \beta_1 x_i$ (the mean follows a line)
2. **Independence:** $\varepsilon_1, \dots, \varepsilon_n$ are independent
3. **Zero mean:** $E[\varepsilon_i] = 0$ for all i
4. **Homoscedasticity:** $\text{Var}(\varepsilon_i) = \sigma^2$ for all i (same variance; **no normality required!**)

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Key point: The Gauss-Markov theorem does **not** require normality.

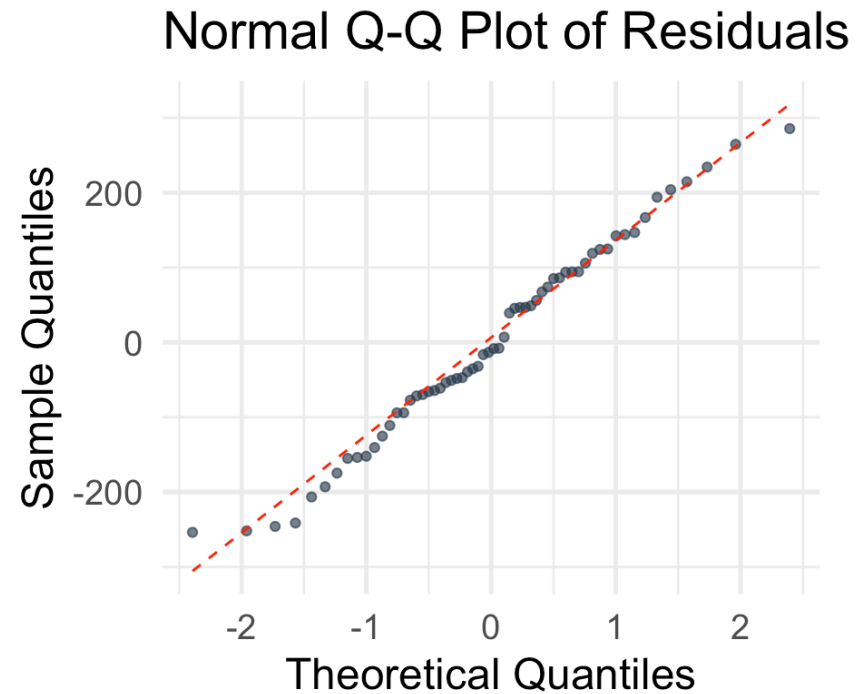
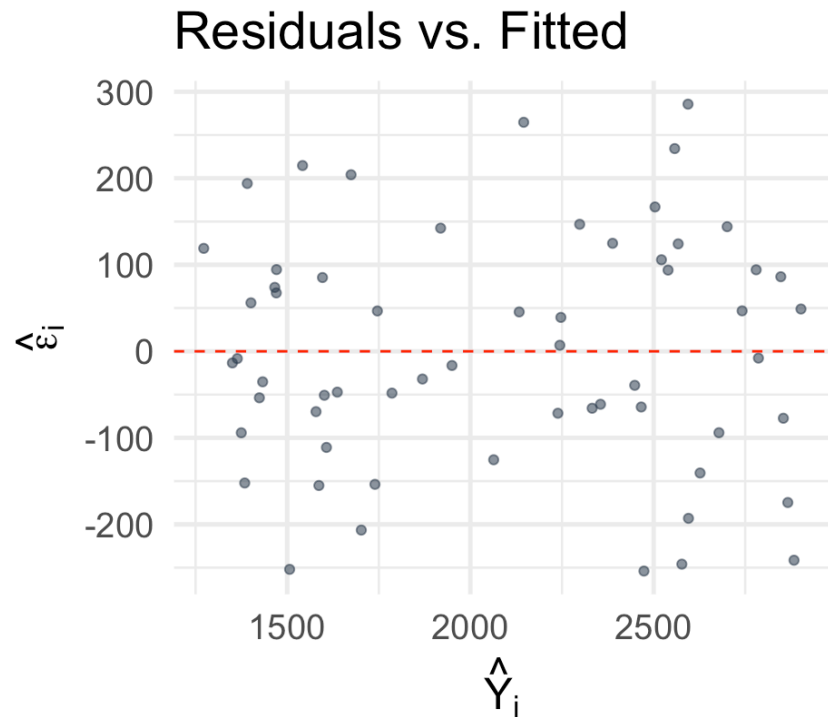
Under these four assumptions alone, OLS is BLUE.

Under the additional assumption $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$, OLS is even stronger: it equals the MLE and is the MVUE, not just BLUE.

Checking the GM Assumptions in Our Example

► Code

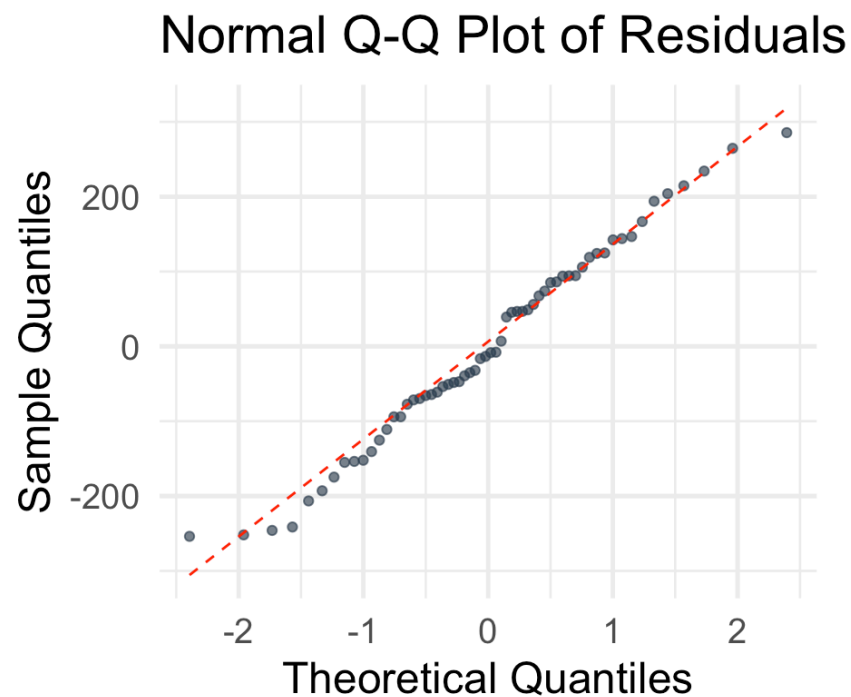
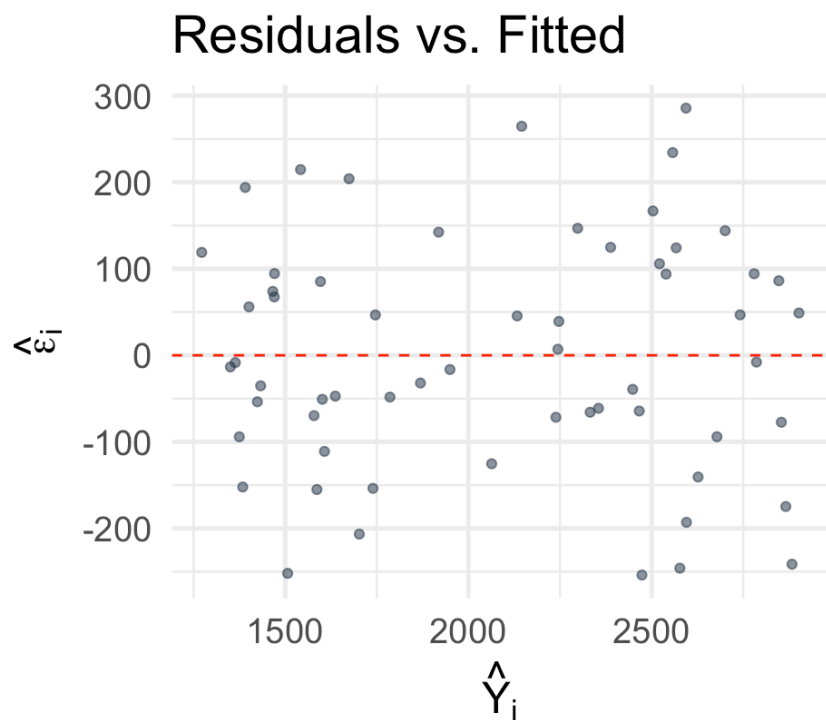
Regression Diagnostics: Birth Weight vs. Gestational Age



Checking the GM Assumptions in Our Example

► Code

Regression Diagnostics: Birth Weight vs. Gestational Age



Residuals look scattered around zero with roughly constant spread; the GM assumptions appear reasonable here.

Part 2: The OLS Estimators: Derivation

The Principle of Least Squares

Question: Given data $(x_1, Y_1), \dots, (x_n, Y_n)$, how do we choose $\hat{\beta}_0$ and $\hat{\beta}_1$?

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! Principle of Least Squares (DBC §12.1)

Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the **residual sum of squares**:

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$$

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This minimizes the total squared vertical distance from each data point to the line.

Why squares? Calculus works cleanly; squared deviations penalize large errors more than small ones; connects to maximum likelihood under normality (see later).

Deriving the OLS Estimates: Normal Equations

Take partial derivatives and set to zero:

$$\frac{\partial \text{RSS}}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) = 0$$

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These simplify to the **normal equations** (DBC §12.1):

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i = \sum Y_i$$

$$\hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2 = \sum x_i Y_i$$

Solving the Normal Equations

From the first equation: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$

Substituting into the second and simplifying:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

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$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

where $S_{xx} = \sum (x_i - \bar{x})^2$ and $S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$.

Computing OLS in Our Example

```
1 # Manual computation of OLS estimators
2 x <- preterm$gestational_age
3 Y <- preterm$birth_weight
4
5 x_bar <- mean(x)
6 Y_bar <- mean(Y)
7
8 Sxx <- sum((x - x_bar)^2)
9 Sxy <- sum((x - x_bar) * (Y - Y_bar))
10
11 beta1_hat <- Sxy / Sxx
12 beta0_hat <- Y_bar - beta1_hat * x_bar
13
14 cat("Manual OLS:\n")
```

Manual OLS:

```
1 cat("  beta0_hat =", round(beta0_hat, 2), "\n")
```

beta0_hat = -3181.73

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1 cat("  beta1_hat =", round(beta1_hat, 2), "\n\n")
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beta1_hat = 169.88

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1 # Confirm with lm()
2 fit <- lm(birth_weight ~ gestational_age, data = preterm)
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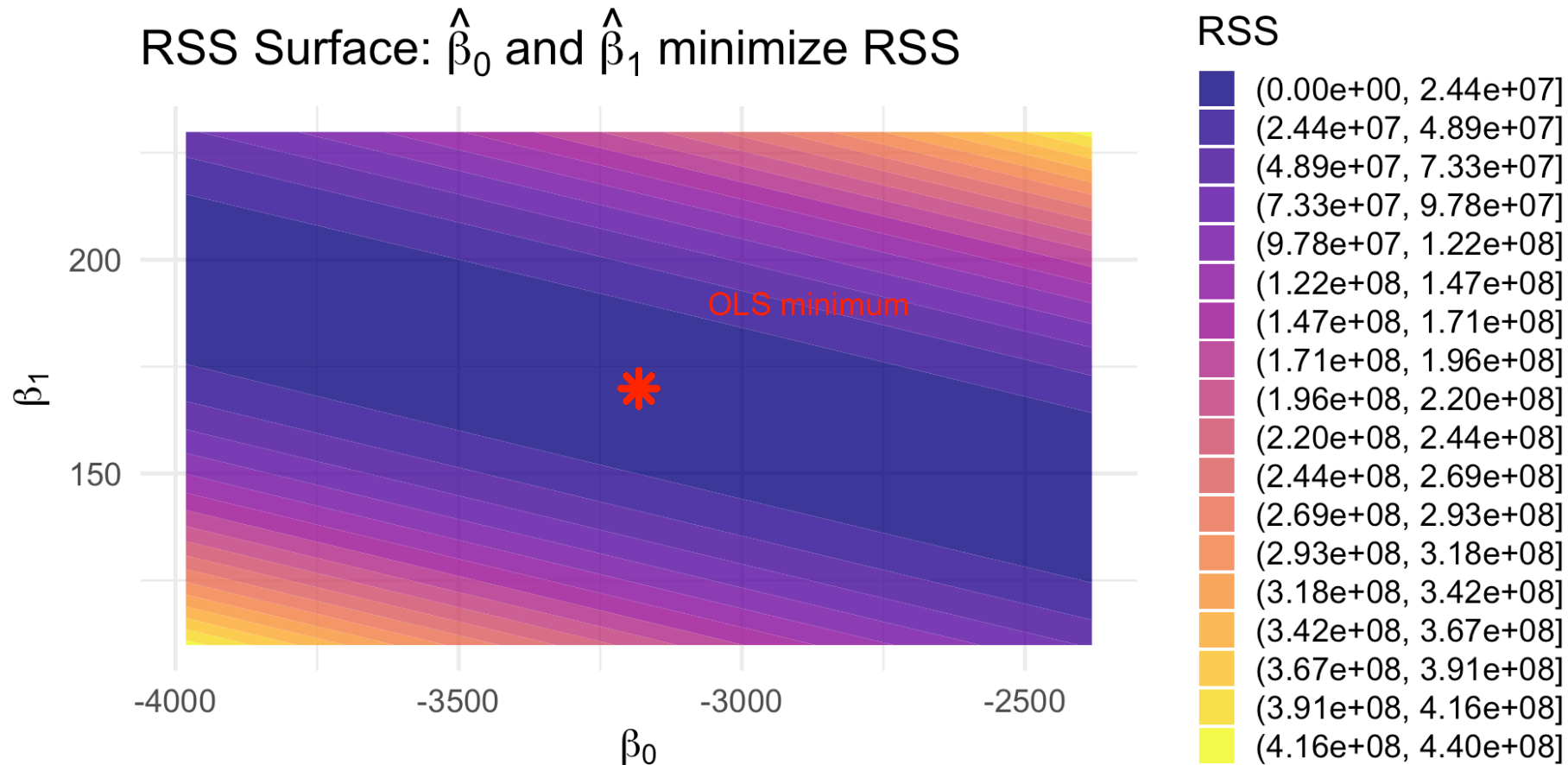
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The RSS Surface: A Geometric View

► Code



The OLS solution is at the **unique global minimum** of the convex RSS surface.



Part 3: Properties of OLS Estimators

Step 1: OLS Estimators Are Linear in Y

Key rewrite: The slope estimator can be written as a linear combination of the Y_i 's:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}} = \sum_{i=1}^n c_i Y_i$$

where $c_i = \frac{x_i - \bar{x}}{S_{xx}}$ are **fixed constants** (they depend only on the x_i 's, not on the Y_i 's).

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Definition: Linear Estimator

An estimator $\tilde{\beta}$ is **linear** if it can be written as $\tilde{\beta} = \sum_{i=1}^n a_i Y_i$ for some constants a_i that depend only on the x_i 's.

OLS estimators satisfy this definition; they are **linear estimators**.

Step 2: OLS Estimators Are Unbiased

Both $\hat{\beta}_0$ and $\hat{\beta}_1$ are **unbiased estimators** of β_0 and β_1 , respectively, under the GM assumptions.

For $\hat{\beta}_1$: Since $\hat{\beta}_1 = \sum c_i Y_i$ and $E[Y_i] = \beta_0 + \beta_1 x_i$:

$$\begin{aligned} E[\hat{\beta}_1] &= E \left[\sum c_i Y_i \right] = \sum c_i E[Y_i] = \sum c_i (\beta_0 + \beta_1 x_i) \\ &= \beta_0 \underbrace{\sum c_i}_{=0} + \beta_1 \underbrace{\sum c_i x_i}_{=1} = \beta_1 \checkmark \end{aligned}$$

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The two facts that make this work, for $c_i = (x_i - \bar{x})/S_{xx}$:

$$\sum_{i=1}^n c_i = \frac{\sum (x_i - \bar{x})}{S_{xx}} = \frac{0}{S_{xx}} = 0$$

because deviations from the mean always sum to zero.

$$\sum_{i=1}^n c_i x_i = \frac{\sum (x_i - \bar{x}) x_i}{S_{xx}} = \frac{\sum (x_i - \bar{x})(x_i - \bar{x}) + \bar{x} \sum (x_i - \bar{x})}{S_{xx}} = \frac{S_{xx} + 0}{S_{xx}} = 1$$

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Step 3: Exact Variances of OLS Estimators

Under the GM assumptions (ε_i independent with $\text{Var}(\varepsilon_i) = \sigma^2$):

$$\text{Var}(\hat{\beta}_1) = \text{Var} \left(\sum c_i Y_i \right) = \sum c_i^2 \text{Var}(Y_i) = \sigma^2 \sum c_i^2 = \frac{\sigma^2}{S_{xx}}$$

$$\text{since } \sum c_i^2 = \sum \frac{(x_i - \bar{x})^2}{S_{xx}^2} = \frac{S_{xx}}{S_{xx}^2} = \frac{1}{S_{xx}}.$$

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! Variance Formulas (DBC §12.2)

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}}$$

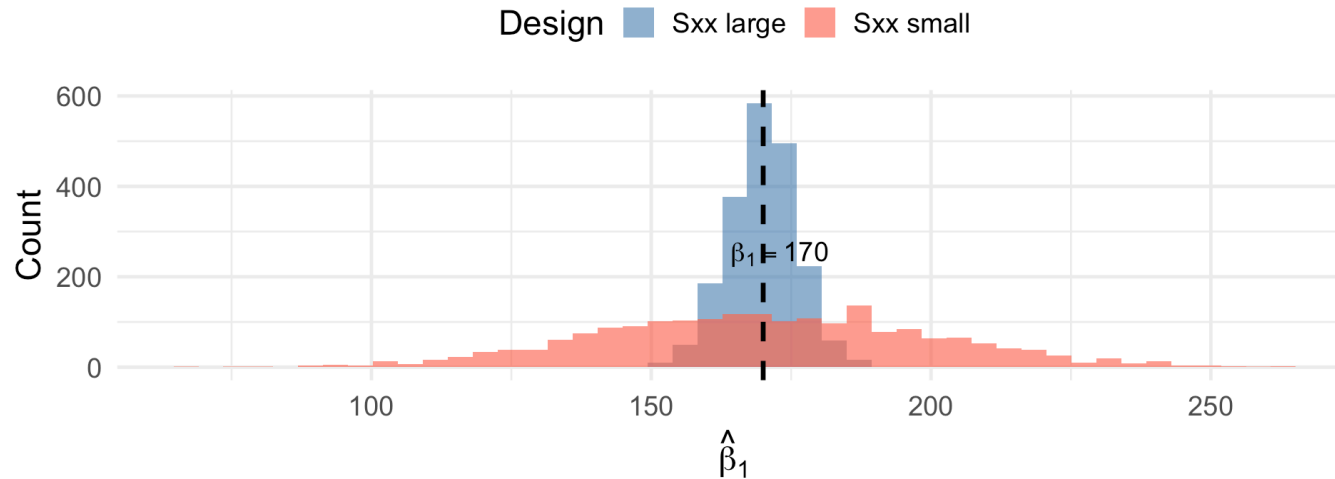
Intuition: What Drives Variance of $\hat{\beta}_1$?

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

► Code

Variance of $\hat{\beta}_1$ depends on spread of x values

$$\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$$



Lesson for study design: Spread your predictor values out to get a more precise slope estimate!

Part 4: The Gauss-Markov Theorem

Statement of the Gauss-Markov Theorem

! Gauss-Markov Theorem (DBC §12.2)

Under the four Gauss-Markov assumptions (linearity, independence, zero-mean errors, homoscedasticity), the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **Best Linear Unbiased Estimators** (BLUE) of β_0 and β_1 .

That is, among all estimators of the form $\tilde{\beta}_1 = \sum a_i Y_i$ that satisfy $E[\tilde{\beta}_1] = \beta_1$:

$$\text{Var}(\hat{\beta}_1) \leq \text{Var}(\tilde{\beta}_1)$$

with equality iff $\tilde{\beta}_1 = \hat{\beta}_1$.

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Unpacking “BLUE”:

- **Best** = smallest variance among the class considered
- **Linear** = estimator is a linear function of the Y_i 's
- **Unbiased** = $E[\tilde{\beta}] = \beta$ for all $\beta_0, \beta_1, \sigma^2$
- **Estimator** = a function of the observed data

Proof: Gauss-Markov for $\hat{\beta}_1$

Setup: Let $\tilde{\beta}_1 = \sum a_i Y_i$ be any linear unbiased estimator of β_1 , where the a_i 's are constants.

Unbiasedness constraint:

$$E[\tilde{\beta}_1] = \sum a_i E[Y_i] = \sum a_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum a_i + \beta_1 \sum a_i x_i = \beta_1$$

This must hold for all β_0 and β_1 , so: $\sum a_i = 0$ and $\sum a_i x_i = 1$.

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Write $a_i = c_i + e_i$ where $c_i = (x_i - \bar{x})/S_{xx}$ are the OLS weights and $e_i = a_i - c_i$.

The constraints $\sum a_i = 0$ and $\sum a_i x_i = 1$ imply $\sum e_i = 0$ and $\sum e_i x_i = 0$.

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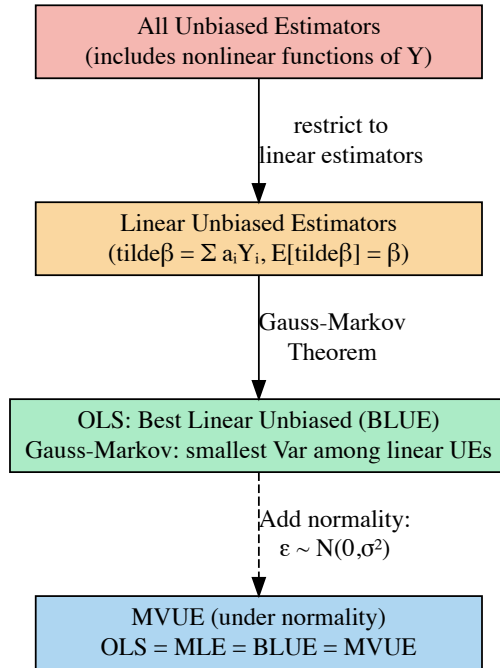
Variance of $\tilde{\beta}_1$:

$$\text{Var}(\tilde{\beta}_1) = \sigma^2 \sum a_i^2 = \sigma^2 \sum (c_i + e_i)^2 = \sigma^2 \left(\sum c_i^2 + 2 \sum c_i e_i + \sum e_i^2 \right)$$

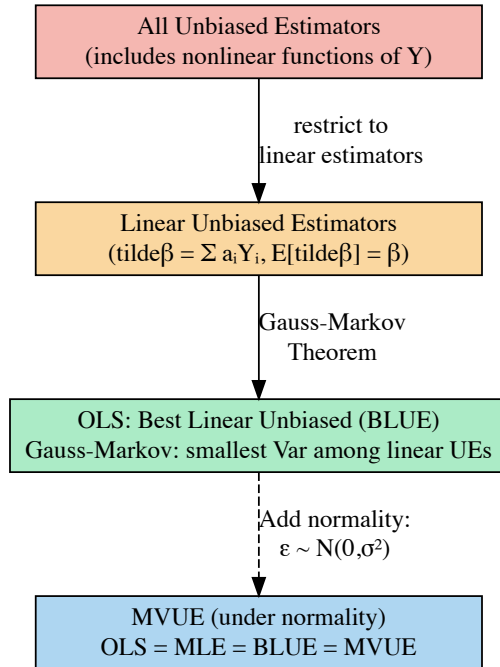
Since $\sum c_i e_i = \frac{1}{S_{xx}} \sum (x_i - \bar{x}) e_i = \frac{1}{S_{xx}} (\sum x_i e_i - \bar{x} \sum e_i) = 0$, we get:

$$\text{Var}(\tilde{\beta}_1) = \underbrace{\frac{\sigma^2}{S_{xx}}}_{\text{Var}(\hat{\beta}_1)} + \underbrace{\sigma^2 \sum e_i^2}_{\geq 0} \geq \text{Var}(\hat{\beta}_1) \quad \checkmark$$

The BLUE Hierarchy: Connecting to MVUE



The BLUE Hierarchy: Connecting to MVUE



- **Gauss-Markov (no normality needed):** OLS is BLUE — smallest variance among all linear unbiased estimators
- **Add normality:** two independent arguments both conclude OLS is MVUE — see next slide

From BLUE to MVUE: Two Routes Under Normality

Adding $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ unlocks two separate arguments, each concluding OLS is the MVUE.

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! Route 1: Lehmann-Scheffé

Under the normal SLR model, $(\hat{\beta}_0, \hat{\beta}_1, S^2)$ is a **complete sufficient statistic** for $(\beta_0, \beta_1, \sigma^2)$.

$\hat{\beta}_1$ is an **unbiased function** of that complete sufficient statistic.

Therefore, by the Lehmann-Scheffé theorem, $\hat{\beta}_1$ is the **unique MVUE** of β_1 .

The CRLB plays no role in this argument — it follows from completeness and sufficiency alone.

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! Route 2: CRLB / Efficiency

The normal SLR model belongs to the **exponential family**. For exponential family models, the MLE achieves the Cramér-Rao Lower Bound **exactly** (not just asymptotically).

Since OLS = MLE under normality (shown in Part 5), and $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$ equals the CRLB:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} = \text{CRLB for } \beta_1$$

An unbiased estimator that achieves the CRLB is, by definition, the MVUE.

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These are parallel routes to the same destination. Lehmann-Scheffé works via the structure of the sufficient statistic; the CRLB route works via the information bound. Both require normality.

What BLUE Does Not Say

Common Misconceptions

BLUE does not mean:

- OLS has the smallest variance among **all** estimators (biased or not): weighted least squares (WLS) or ridge regression can have smaller MSE
- OLS is optimal when GM assumptions are **violated**: heteroscedasticity calls for WLS; dependence calls for generalized least squares (GLS)
- The model is **correct**: OLS is BLUE for the stated model, regardless of whether the model fits the data well

BLUE does mean:

- Within the class of **linear, unbiased** estimators, OLS cannot be beaten
- This is a remarkable, assumption-lean result: no distributional assumption on ε_i beyond mean 0 and constant variance

Part 5: OLS = MLE Under Normality

The Normal Simple Linear Regression Model

Add one assumption to the GM setup:

$$\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

This implies:

$$Y_i \stackrel{\text{indep}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

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The joint likelihood is:

$$\begin{aligned} L(\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(Y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\} \\ &= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2\right\} \end{aligned}$$

Maximizing the Likelihood

The log-likelihood is:

$$\ell(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$$

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Key observation: For fixed σ^2 , maximizing ℓ with respect to β_0 and β_1 is **equivalent to minimizing**:

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! OLS = MLE Under Normality

The OLS estimators $\hat{\beta}_0^{\text{OLS}}$ and $\hat{\beta}_1^{\text{OLS}}$ are **identical** to the MLE estimators $\hat{\beta}_0^{\text{MLE}}$ and $\hat{\beta}_1^{\text{MLE}}$.

This is why we can use t -tests and F -tests in normal linear regression; the distributional framework that enables inference comes from normality + OLS = MLE.

From BLUE to MVUE: The Full Chain

Under the normal SLR model ($\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$):

Property	Justification
$\hat{\beta}_1$ is linear	By construction: $\hat{\beta}_1 = \sum c_i Y_i$
$\hat{\beta}_1$ is unbiased	Shown above (only needs GM assumptions)
$\hat{\beta}_1$ is BLUE	Gauss-Markov Theorem (only needs GM assumptions)
$\hat{\beta}_1$ is sufficient	$(\hat{\beta}_0, \hat{\beta}_1, S^2)$ is sufficient for $(\beta_0, \beta_1, \sigma^2)$ under normality
$\hat{\beta}_1$ is the MLE	OLS = MLE under normality (shown above)
$\hat{\beta}_1$ is the MVUE	Lehmann-Scheffé: complete sufficient statistic + unbiased
$\hat{\beta}_1$ achieves CRLB	MLE achieves CRLB asymptotically; exactly for exponential family

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 The Big Picture

This is precisely the situation we studied with scalar parameters: the MLE of an exponential family parameter based on the sufficient statistic is unbiased, complete, and achieves the CRLB. The normal SLR model is an exponential family model, and OLS produces the sufficient statistics.

Connecting to the CRLB

Under normal SLR, the Fisher information matrix for (β_0, β_1) gives:

$$\text{CRLB for } \beta_1 = \frac{\sigma^2}{S_{xx}} = \text{Var}(\hat{\beta}_1^{\text{OLS}})$$

$$\text{CRLB for } \beta_0 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) = \text{Var}(\hat{\beta}_0^{\text{OLS}})$$

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OLS estimators **achieve the Cramér-Rao Lower Bound**; they are **efficient** unbiased estimators.

```
1 # Verify: compare theoretical variance to simulation variance
2 sigma <- 120
3 Sxx_val <- Sxx
4
5 cat("Theoretical Var(beta1_hat) =", round(sigma^2 / Sxx_val, 4), "\n")
```

Theoretical Var(beta1_hat) = 25.1581

```
1 cat("Simulated Var(beta1_hat) =", round(var(slopes_high), 4), "\n")
```

Simulated Var(beta1_hat) = 36.8917

Part 6: R Simulation: Verifying Gauss-Markov

Comparing OLS to Alternative Linear Estimators

The GM theorem says OLS beats every other linear unbiased estimator. Let's verify with a simulation.

Setting: Same premature infant model: $Y_i = -3200 + 170x_i + \varepsilon_i, \varepsilon_i \sim N(0, 120^2)$

Competitors:

1. **OLS:** $\hat{\beta}_1 = S_{xy}/S_{xx}$ (the Gauss-Markov champion)
2. **Naïve two-point slope:** Use only the first and last observation: $\tilde{\beta}_1^{(2)} = (Y_n - Y_1)/(x_n - x_1)$
3. **Trimmed OLS:** Drop middle third of observations, apply OLS to rest
4. **Unweighted mean of slopes:** Average of all pairwise slopes (all pairs $i < j$)

All four are linear in Y and unbiased, but only OLS is BLUE.

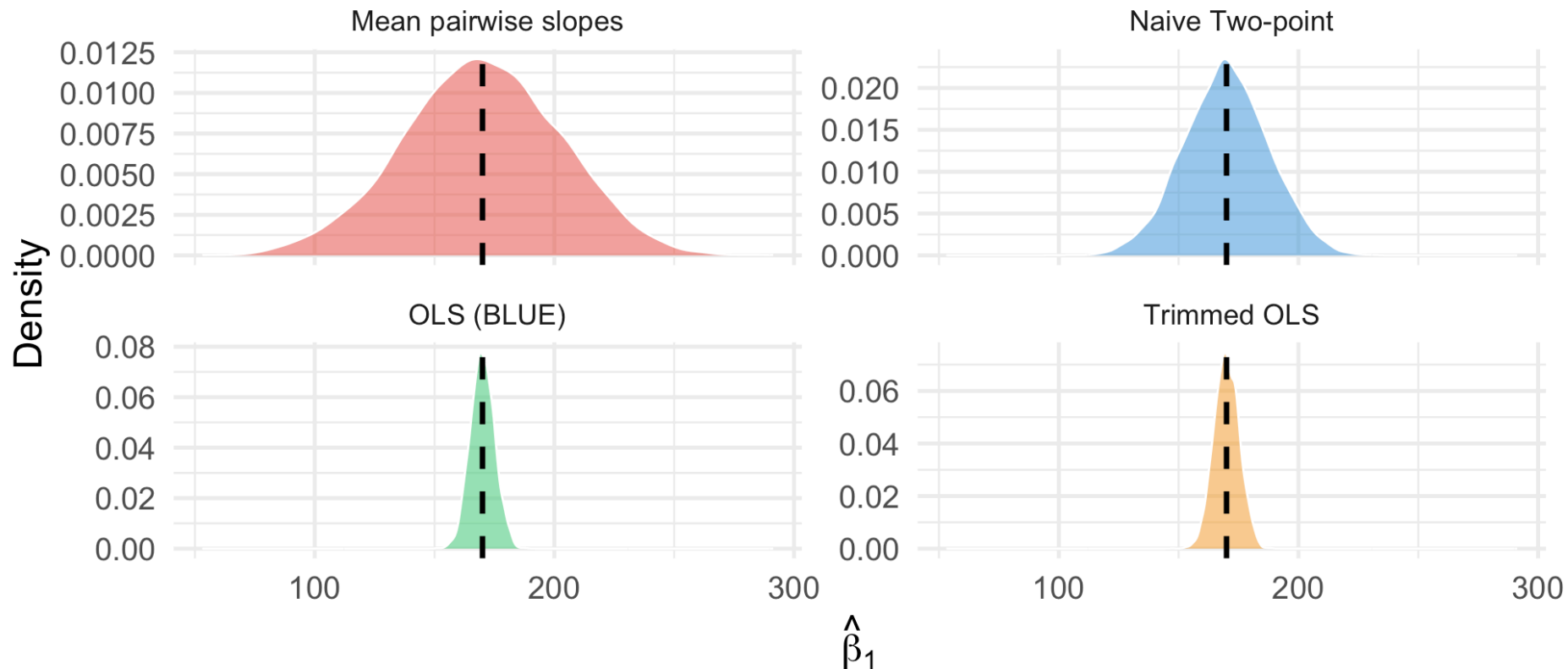
► Code

Simulation Results: OLS Wins

► Code

Sampling Distributions of Four Linear Unbiased Estimators of β_1

True $\beta_1 = 170$ (dashed line); OLS has smallest variance



Summarizing the Simulation

► Code

Comparison of Linear Unbiased Estimators (5,000 simulations)

Estimator	Mean	Variance	SD	Relative Efficiency
OLS (BLUE)	169.9583	25.4524	5.0450	1.000
Trimmed OLS	169.9690	27.3717	5.2318	0.930
Two-point	169.8514	312.1433	17.6676	0.082
Mean pairwise	169.8039	1062.4175	32.5947	0.024

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► Code

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Interpretation of relative efficiency: The ratio of OLS variance to each competitor's variance; OLS (efficiency = 1.0) is the benchmark.

Part 7: Estimation of σ^2 and Inference

Estimating the Error Variance σ^2

The GM assumptions include an unknown error variance σ^2 . We need to estimate it for inference.

Residuals: $\hat{\varepsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = Y_i - \hat{Y}_i$

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! Unbiased Estimator of σ^2 (DBC §12.2)

$$\hat{\sigma}^2 = S^2 = \frac{\text{RSS}}{n-2} = \frac{\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{n-2}$$

$E[S^2] = \sigma^2$. Why $n - 2$? We lose 2 degrees of freedom because we estimate 2 parameters.

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$E[S^2] = \sigma^2$. Why $n-2$? We lose 2 degrees of freedom because we estimate 2 parameters.

Under normality: $\frac{(n-2)S^2}{\sigma^2} \sim \chi_{n-2}^2$, independent of $\hat{\beta}_0$ and $\hat{\beta}_1$.

This is the direct analog of how S^2 worked in one-sample inference: the distribution changes because the number of estimated parameters changes.

Standard Errors and t-Based Inference

Replacing σ^2 with S^2 gives estimated standard errors:

$$\widehat{\text{SE}}(\hat{\beta}_1) = \frac{S}{\sqrt{S_{xx}}} \quad \widehat{\text{SE}}(\hat{\beta}_0) = S \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

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Under normality, the pivot statistics follow t distributions:

$$T_1 = \frac{\hat{\beta}_1 - \beta_1}{S/\sqrt{S_{xx}}} \sim t_{n-2} \quad T_0 = \frac{\hat{\beta}_0 - \beta_0}{S\sqrt{1/n + \bar{x}^2/S_{xx}}} \sim t_{n-2}$$

```
1 # Standard errors in our example
2 fit_summary <- summary(fit)
3 cat("sigma_hat (S) =", round(fit_summary$sigma, 2), "\n")
```

sigma_hat (S) = 136.42

```
1 tidy(fit) |> select(term, estimate, std.error, statistic, p.value) |>
2 mutate(p.value = scales::pvalue_format(0.0001)(p.value), across(where(is.numeric), \(x) r
```

A tibble: 2 × 5

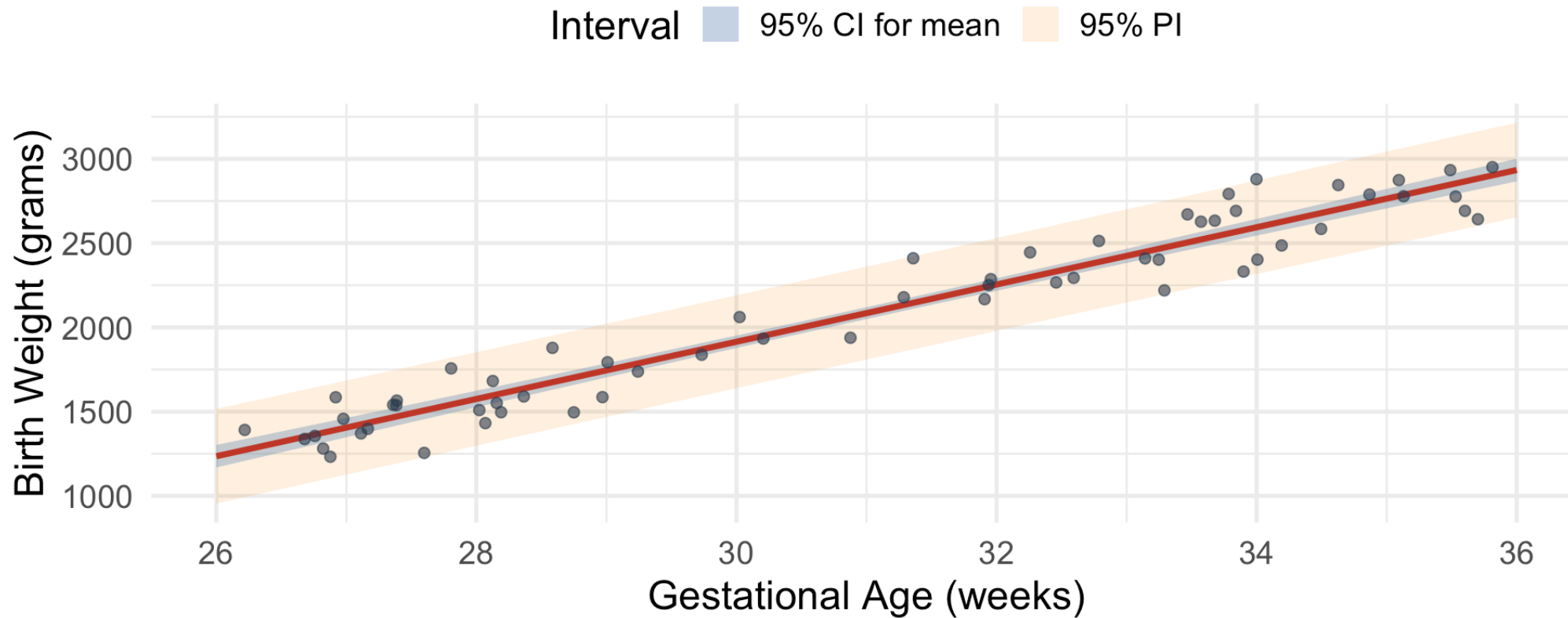
term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<chr>

Prediction and the Fitted Line

► Code

Birth Weight vs. Gestational Age: Fitted Line with Uncertainty Bands

CI for $E[Y | x]$, PI for an individual future Y



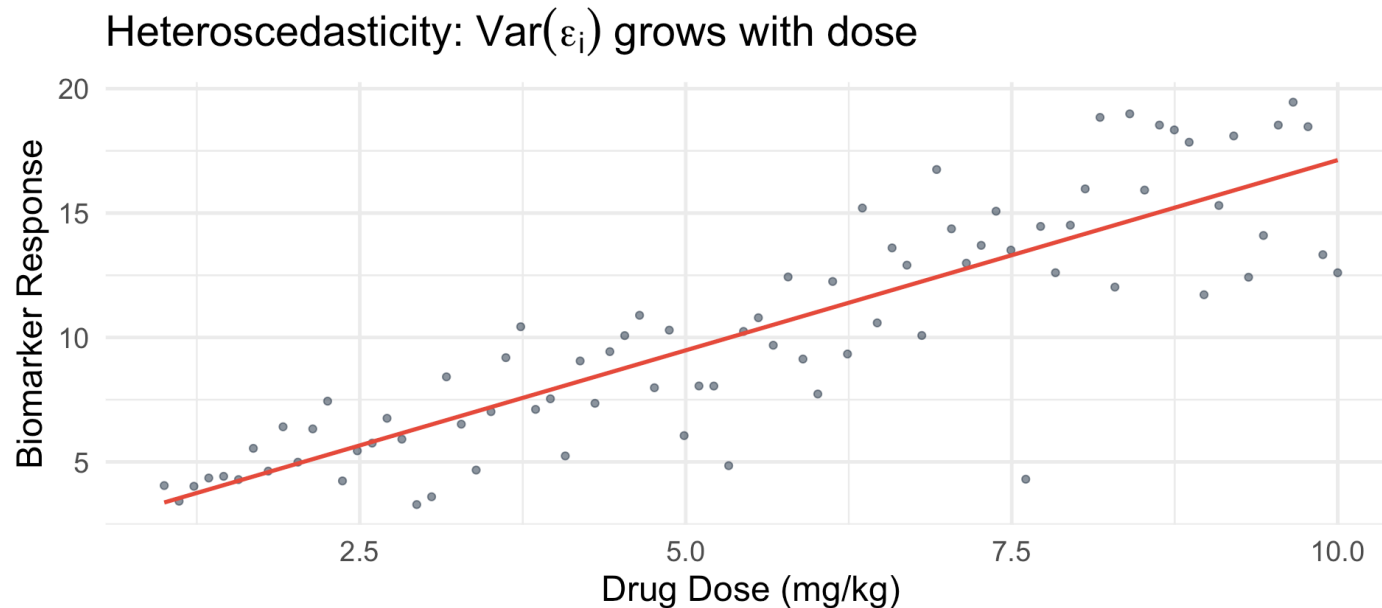
Part 8: When Gauss-Markov Fails

Violation: Heteroscedasticity

If $\text{Var}(\varepsilon_i) = \sigma_i^2$ (not constant), OLS is still **unbiased** but is no longer BLUE.

Example: In a study of drug dosage and response, variance in response may increase with dose.

► Code

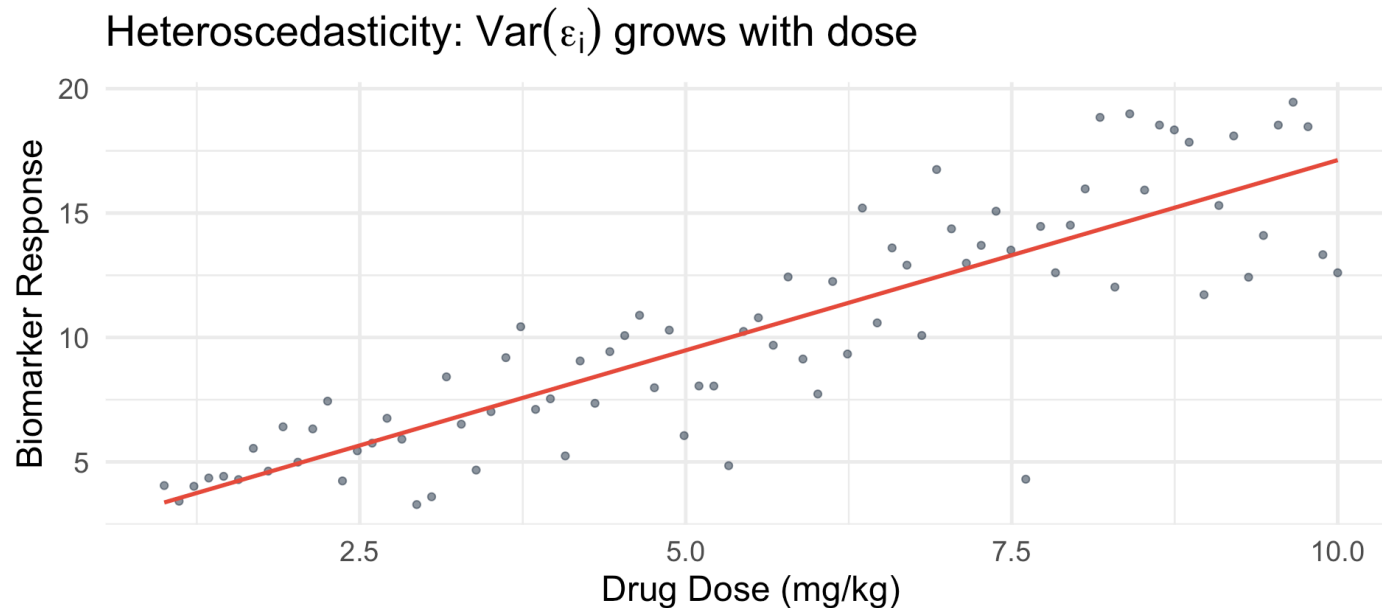


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► Code



Fix: Weighted Least Squares (WLS) minimizes $\sum w_i(Y_i - \beta_0 - \beta_1 x_i)^2$ where $w_i \propto 1/\sigma_i^2$, down-weighting observations with higher variance.

Weighted least squares is BLUE when weights are correctly specified. This is DBC §12.9.

Summary: The Hierarchy of OLS Optimality

Assumption Set	OLS Property
GM 1-4 (linearity, independence, $E[\varepsilon_i] = 0$, homoscedasticity)	BLUE : best among linear unbiased estimators
GM 1-4 + normality ($\varepsilon_i \sim N(0, \sigma^2)$)	MVUE : best among all unbiased estimators
GM 1-4 + normality	MLE : OLS and MLE coincide
GM 1-4 + normality	Achieves CRLB : efficient
Homoscedasticity violated	OLS unbiased, but not BLUE; use weighted least squares (WLS)
Independence violated	OLS unbiased, but not BLUE; use generalized least squares (GLS)

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Homoscedasticity violated	OLS unbiased, but not BLUE; use weighted least squares (WLS)
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The Takeaway for Applied Regression

When you check regression diagnostics (residual plots, Q-Q plots, leverage), you are **verifying the GM assumptions** that justify the optimality of OLS. If they fail, you need a different estimator.

Part 9: Extended Example from DBC

DBC Example 12.7: Tensile Strength and Percentage Cotton

From DBC §12.1–12.2: Tensile strength (psi) of synthetic fiber as a function of percentage of cotton in the fiber.

► Code

```
Sxx = 1000    Sxy = 160
```

► Code

```
beta1_hat = 0.16
```

► Code

```
beta0_hat = 10.8
```

► Code

```
Theoretical Var(beta1_hat) / sigma^2 = 0.001
```

► Code

```
# A tibble: 2 × 5
```

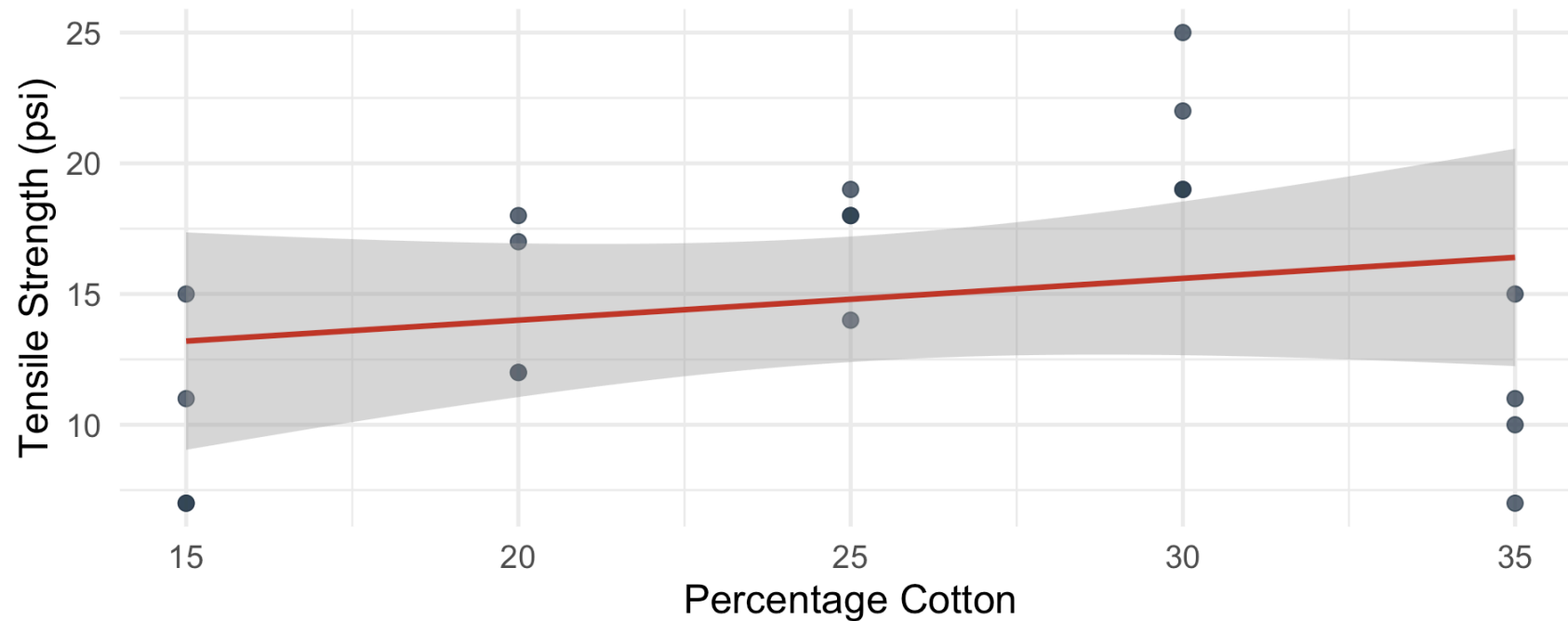
	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	10.8	4.20	2.57	0.0191
2	pct_cotton	0.16	0.162	0.991	0.335

DBC Example 12.7: Plot

► Code

DBC Example 12.7: Tensile Strength vs. Cotton Content

$\hat{\beta}_1 = 0.16$ psi per % cotton (OLS = BLUE = MLE)

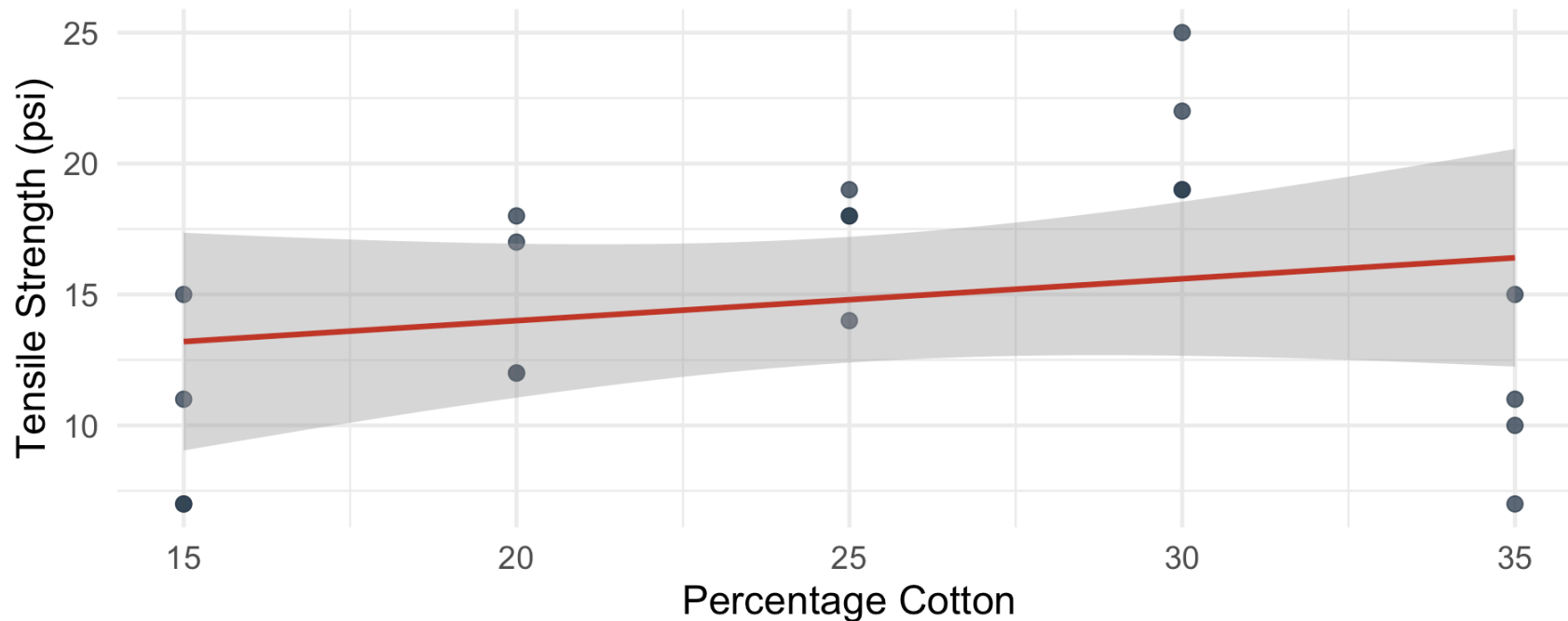


DBC Example 12.7: Plot

► Code

DBC Example 12.7: Tensile Strength vs. Cotton Content

$\hat{\beta}_1 = 0.16$ psi per % cotton (OLS = BLUE = MLE)



The positive slope $\hat{\beta}_1 \approx 0.16$ is the **BLUE** of the true slope. Under normality, it also achieves the CRLB and is the MVUE.

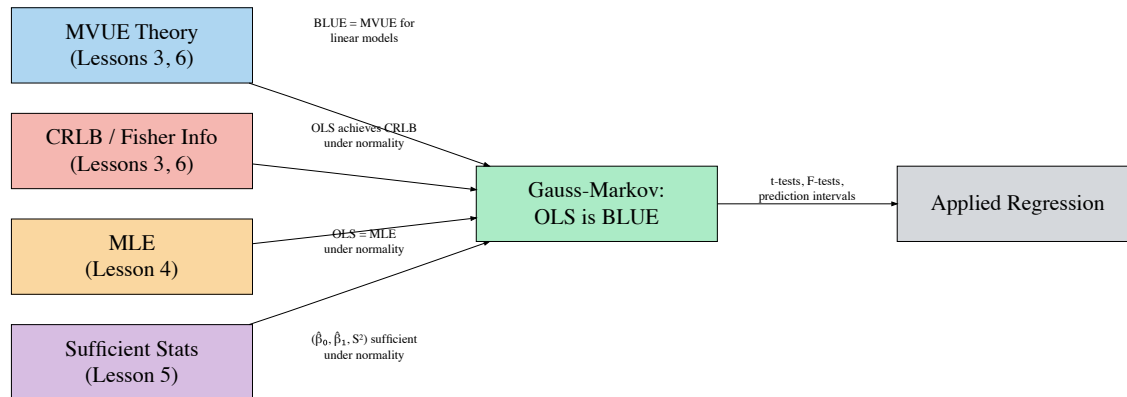
Summary

What We Proved Today

! Key Results

1. **OLS estimators** $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$ minimize the RSS
2. OLS estimators are **linear** in Y_i and **unbiased** under GM assumptions 1–3
3. Under all four GM assumptions: $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$, $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + \bar{x}^2/S_{xx})$
4. **Gauss-Markov Theorem**: OLS is **BLUE** (no normality required)
5. Under normality: OLS = MLE, and OLS is the **MVUE** (not just BLUE)
6. OLS achieves the **Cramér-Rao Lower Bound**; it is an **efficient** estimator under normality

Connections to the Course Themes



Looking Ahead

- **Applied Regression:** Everything builds on today's theory
 - Multiple regression extends to $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$; the same BLUE result holds via matrix algebra
 - Generalized Linear Models (GLMs): when ε_i are non-normal $\rightarrow$ MLE replaces OLS
 - Mixed models: when independence fails $\rightarrow$ GLS / REML
- **Weighted Least Squares:** BLUE when heteroscedasticity is known (DBC §12.9)

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 - Mixed models: when independence fails $\rightarrow$ GLS / REML
- **Weighted Least Squares:** BLUE when heteroscedasticity is known (DBC §12.9)

Matrix Formulation Preview

In multiple regression: $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$

OLS estimate: $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

This is still linear in $\mathbf{Y}$, still unbiased under GM conditions, and still BLUE by the matrix version of the Gauss-Markov theorem.

Questions?

Thank you!